

Abhishek Singh

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Education

Indian Institute of Technology Patna

2023–2027 (expected)

B.Tech in Electronics and Communication Engineering

CPI: 7.00/10

Relevant coursework (completed / ongoing): Microcontroller and Embedded System; Microprocessor; Digital Electronics; VLSI Design; Control Systems; Signals and Systems; Digital Signal Processing; Internet of Things; Computer Organization and Architecture; Automotive Technology.

Technical Skills

Embedded systems: ESP32, ADC-based signal acquisition, I2S, Bluetooth Serial Port Profile (SPP)

Programming: C, C++, Python; NumPy for numerical analysis and validation

Simulation and analysis: MEEP (FDTD), MPB; parameter sweeps, noise modeling, signal processing

Development tools: Git, Linux

Hardware & Embedded Projects

ESP32 Smart Stethoscope

Embedded signal acquisition

- Developed an ESP32-based audio acquisition prototype using a MAX9814 microphone amplifier and ADC1_CH0 input, with capacitive coupling in the analog signal path.
- Implemented 44.1 kHz, 16-bit mono acquisition using I2S, with a 20–250 Hz bandpass filter for the acquired signal.
- Integrated Bluetooth SPP streaming with a Cordova mobile application for audio reception.

Prabha / PEMA: Photonic–Electronic Computing

IIT Patna | May 2026–Present

Research under Prof. Sumanta Gupta, Photonics / Optical Communication Group

- Developed modular Python models of a continuous-wave laser and an ideal differential photonic multiply–accumulate (MAC) block for signed-weight computation.
- Modeled laser relative intensity noise and linewidth-induced phase diffusion; validated power, wavelength, sampling-rate and random-seed behavior through parameter sweeps.
- Recovered a modeled 1 MHz linewidth within approximately 5.1% using an ensemble of 1,000 realizations; verified ideal MAC output across 0.1, 1 and 10 mW input powers.

Silicon Photonic Microring Resonator Simulation

May–June 2026

- Simulated lateral, vertical and cascaded microring coupling configurations using MEEP FDTD; studied coupling-gap dependence, resonance and quality factor.
- Used MPB to characterize optical modes and effective indices; examined numerical convergence to assess simulation reliability.

Additional Experience

Software Head — Inditronix AI Labs

April 2025–May 2026

- Coordinated a multidisciplinary software team for healthcare AI solutions and developed ESP32-based IoT applications.

Video Surveillance Research Intern — IIT Patna

May–July 2026

Under Prof. Mahesh Kumar Kolekar

- Developed a 3D convolutional autoencoder and hybrid patch-feature scoring pipeline for video anomaly detection; achieved ROC-AUC of 0.734 on UCSD Ped2.

Achievements & Affiliations

- GATE ECE 2026: AIR 1046; score 662.** JEE Advanced: AIR 5701; JEE Main: AIR 4311.
- Competitive programming:** Codeforces Pupil (peak 1241); CodeChef 3-star (peak 1643).
- IEEE Student Member, IEEE Photonics Society (since June 2026).